

textGrain: Entropy-Calibrated Watermarking for Language Model Text

Technical report accompanying OpenAI’s October 5th blog post

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Abstract

This technical report provides the mathematical details of the textGrain watermark described in OpenAI’s October 5th blog post. It specifies the generation and detection procedures, explains how to set a budget for the sampling entropy lost in exchange for a watermark signal, and states the assumptions underlying the theoretical guarantees. The method couples token generation to keyed randomness through an optimal transport problem with costs based on Gumbel random variables and Kullback–Leibler (KL) regularization. The KL divergence from independence equals the reduction in conditional entropy, giving the watermark strength parameter a direct information-theoretic interpretation. To speed up computation, we apply transport to vocabulary blocks, reducing the optimization dimension while preserving relative token probabilities within each block. We analyze how to calibrate the watermark to a chosen entropy-loss budget. The detector requires only the generated text and the secret key and does not need to know the budget used during generation.

1 Introduction

This report is intended for readers of the accompanying blog post who want a precise account of how textGrain works. It explains how generation creates dependence on keyed randomness, what the entropy budget controls, and how the detector obtains statistical evidence from the text.

A language model generates text one token at a time, sampling each token (a word or part of a word) from a next token prediction (NTP) distribution determined by the preceding text. Watermarking embeds a statistical signal by modifying this sampling using pseudorandom values derived from a secret key and the preceding context (Kirchenbauer et al., 2023; Aaronson, 2023; Dathathri et al., 2024). With the watermark configuration specified, a detector uses the text and key to reconstruct these values and tests for dependence between them and the observed tokens (Li et al., 2025a).

Preserving the NTP distribution provides one way to limit the effect of watermarking on generation. A watermark is *unbiased* if, at any fixed context, averaging its token distribution over

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a randomly drawn key recovers the original NTP distribution (Hu et al., 2024; Li et al., 2025a; Dathathri et al., 2024). In practice, a key is often reused across responses. Unbiasedness alone does not ensure that sampling remains random under that key. For example, Gumbel-max watermarking selects the token with the largest sum of its log probability and a keyed Gumbel value; at a fixed context and key, it always selects the same token (Aaronson, 2023). Repeated generation from the same prompt can therefore produce identical responses (Kuditipudi et al., 2024; He et al., 2025b). This matters when an application needs several distinct answers or solutions to the same prompt.

If tokens are independent of the keyed randomness, there is no watermark signal to detect. If that randomness determines every token, no further sampling randomness remains. Entropy reduction measures the average sampling randomness removed between these extremes: it compares the NTP entropy with the entropy that remains after conditioning on the keyed randomness. An unbiased watermark defines a *coupling*, a joint distribution of the generated token and keyed randomness with prescribed marginals (Li et al., 2025a; Xie et al., 2025; Long et al., 2025; Tsur et al., 2025; He et al., 2025a; Gloaguen et al., 2026). The average entropy reduction is the mutual information between these variables. Optimal transport (OT) provides a way to choose the coupling: its cost favors selected forms of dependence, while KL regularization penalizes departure from independence. The KL penalty equals the average entropy reduction, so an entropy budget directly limits the average sampling randomness that may be removed.

textGrain applies OT to keyed blocks of vocabulary tokens. The coupling determines how the selected block depends on the keyed randomness; within that block, tokens retain their original relative probabilities. This reduces the optimization dimension and preserves unbiasedness when the coupling has the required marginals. Section 2 develops the coupling and entropy formulation, and Section 3 specifies block generation and detection. The appendices give the numerical solver and proofs (Appendix A), compatibility with speculative sampling (Appendix B), and connections to related work (Appendix C).

2 Framework: watermarking as OT

Sampling and coupling. Given a token sequence $w_{1:n}$ over a vocabulary $\mathcal{V}$, let $\mathbf{P}_t$ denote the NTP distribution determined by the prefix $w_{1:t-1}$. At position t , the watermark uses a pseudorandom value Ξ_t derived from the secret key and a window of up to L_{ctx} preceding tokens. For a single position, we suppress t and write $\mathbf{P}$, Ξ , and W for the NTP distribution, keyed randomness, and sampled token. At a fixed prefix, a randomly drawn key induces a distribution μ over Ξ . Averaging over keys means averaging over $\Xi \sim \mu$; fixing the keyed randomness means conditioning on Ξ .

A watermark is unbiased if averaging its conditional token distribution over $\Xi \sim \mu$ recovers $\mathbf{P}$. Thus the joint distribution of (W, Ξ) has marginals $\mathbf{P}$ and μ . We call this joint distribution a *coupling* π and write $\Pi(\mathbf{P}, \mu)$ for the set of all such couplings. Once $\Xi = \xi$ is fixed, the sampling distribution is $\mathbf{Q}(\cdot \mid \xi) = \pi(\cdot \mid \Xi = \xi)$. Unwatermarked sampling corresponds to the independent coupling $\mathbf{P} \otimes \mu$, for which $\mathbf{Q}(\cdot \mid \xi) = \mathbf{P}$ for every ξ (Li et al., 2025a). A watermark makes $\mathbf{Q}(\cdot \mid \xi)$ depend on ξ while preserving its average over μ .

Regularized OT. OT selects a coupling by minimizing its expected cost subject to fixed marginals (Peyré and Cuturi, 2019). We assign each token and randomness pair a cost $c(w, \xi)$, with smaller values for pairs favored by the watermark. To limit dependence, we penalize departure

from the independent coupling using the KL divergence (Kullback and Leibler, 1951; Cover and Thomas, 2006),

$$D_{\text{KL}}(\pi \| \mathbf{P} \otimes \mu) = \int \log \left(\frac{d\pi}{d(\mathbf{P} \otimes \mu)} \right) d\pi.$$

All logarithms are natural. With regularization strength $\lambda > 0$, we choose

$$\pi_\lambda \in \arg \min_{\pi \in \Pi(\mathbf{P}, \mu)} \{ \mathbb{E}_\pi c(W, \Xi) + \lambda D_{\text{KL}}(\pi \| \mathbf{P} \otimes \mu) \}. \quad (1)$$

The cost favors the selected dependence, while the KL penalty limits it. The marginal constraints preserve the NTP distribution after averaging over keys.

Information and entropy. The coupling determines how much sampling randomness remains when the keyed randomness is fixed. The NTP entropy is $H(\mathbf{P}) = -\sum_{w \in \mathcal{V}} P_w \log P_w$, with $0 \log 0 = 0$. Given $\Xi = \xi$, the remaining sampling entropy is $H\{\mathbf{Q}(\cdot | \xi)\}$. Averaging this quantity over Ξ gives the conditional entropy $H_\pi(W | \Xi)$. For any coupling with marginals $\mathbf{P}$ and μ (Cover and Thomas, 2006),

$$D_{\text{KL}}(\pi \| \mathbf{P} \otimes \mu) = I_\pi(W; \Xi) = H(\mathbf{P}) - \mathbb{E}_\Xi H\{\mathbf{Q}(\cdot | \Xi)\}.$$

The KL penalty is therefore exactly the average entropy removed by watermarking. This identity concerns an average over the keyed randomness; it does not impose the same entropy loss for every fixed key.

3 textGrain: watermarking with block OT

3.1 Watermark embedding

We apply OT to blocks of tokens to reduce the optimization dimension while retaining relative token probabilities within each block. An entropy budget limits the average loss of sampling entropy to a fraction $\beta \in [0, 1]$ of the NTP entropy. Figure 1 illustrates the construction: partition the vocabulary, construct a coupling under the budget, then sample a block and a token. Algorithm 1 summarizes one generation step.

Vocabulary blocks. Let $\mathbf{P}$ be the NTP distribution after any temperature scaling or top- p truncation (Holtzman et al., 2020), and let $\mathcal{A} = \{w : P_w > 0\}$ be its support. For a chosen number of blocks B , write $[B] = \{1, \dots, B\}$. A keyed partition $g : \mathcal{V} \rightarrow [B]$ assigns each token to a block. The partition depends only on the key and context window, so the detector can reconstruct it without access to $\mathbf{P}$. The block probabilities are

$$P_b^{\text{blk}} = \sum_{w \in \mathcal{A}: g(w)=b} P_w, \quad b \in [B]. \quad (2)$$

Coupling under an entropy budget. We couple the block distribution $\mathbf{P}^{\text{blk}}$ to m equally likely columns. A coupling π assigns probability $\pi(b, j)$ to the pair (b, j) , with row sums P_b^{blk} and column sums $1/m$. Selecting a column uniformly and then sampling a block conditional on that column therefore recovers $\mathbf{P}^{\text{blk}}$ on average.

For each pair (b, j) , the key and context window generate a standard Gumbel variable Z_{bj} with distribution function $F_G(z) = \exp\{-\exp(-z)\}$. We standardize it as $G_{bj} = (Z_{bj} - \gamma)/(\pi/\sqrt{6})$, where

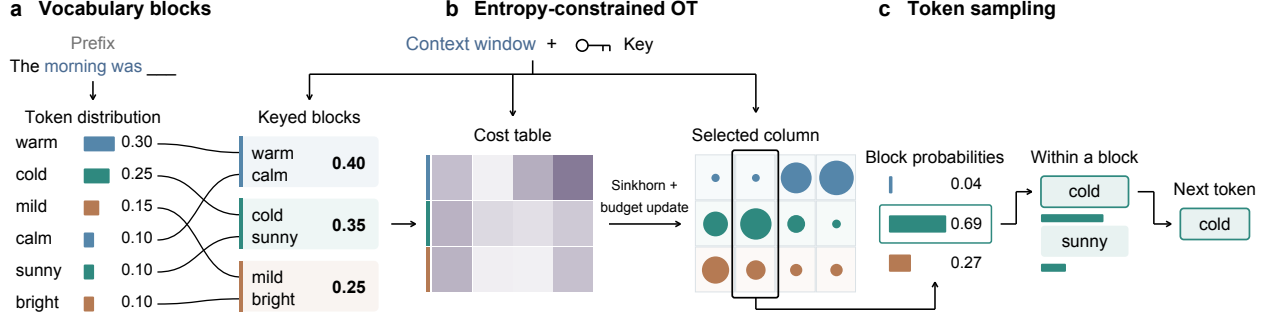


Figure 1: **Embedding a watermark with block OT.** **a**, The key and context window divide the vocabulary into blocks; each block receives the sum of its tokens’ NTP probabilities. Colors distinguish the three blocks in this example. **b**, OT couples these probabilities to four equally likely columns, favoring pairs with smaller costs subject to the entropy budget. The cost table assigns a pseudorandom cost to each pair of a block and a column; darker cells indicate smaller costs. Circle areas show the coupling probabilities. **c**, The key and context window select a column. Its normalized entries determine the block probabilities; a token is then sampled within the selected block using the original relative probabilities. The blue tokens illustrate a context window of two tokens, whereas the model uses the full prefix. Displayed probabilities are rounded.

Algorithm 1 Embedding a watermark with block OT and an entropy budget

Require: NTP distribution $\mathbf{P}$, secret key, context window, block count B , column count m , and target entropy loss fraction β

- 1: Construct the keyed partition g
 - 2: Aggregate the block probabilities using (2)
 - 3: Generate the cost table $c(b, j) = -G_{bj}$ from standardized Gumbel variables
 - 4: Compute a coupling with the prescribed marginals, targeting loss $\beta H(\mathbf{P})$ using Algorithm A.1; retain its achieved loss and convergence status
 - 5: Select the keyed column J , then sample a block and token using (5)
-

γ is the Euler–Mascheroni constant and π here is the mathematical constant. Set $c(b, j) = -G_{bj}$, so minimizing cost favors larger Gumbel values. In the idealized construction, the Z_{bj} are independent and identically distributed, and the random draws used for the partition, cost table, and selected column are independent at a fixed context. Distinct contexts use independent draws. These assumptions model the keyed pseudorandom functions; repeated contexts reuse the same values.

Let $\mathbf{u}_m = (1/m, \dots, 1/m)$. Because relative probabilities within each block are unchanged, the KL divergence from $\mathbf{P}^{\text{blk}} \otimes \mathbf{u}_m$ equals the average loss of NTP entropy across columns. The ideal budgeted problem is

$$\min_{\pi \in \Pi(\mathbf{P}^{\text{blk}}, \mathbf{u}_m)} \sum_{b=1}^B \sum_{j=1}^m \pi(b, j) c(b, j) \quad \text{subject to} \quad D_{\text{KL}}(\pi \| \mathbf{P}^{\text{blk}} \otimes \mathbf{u}_m) \leq \beta H(\mathbf{P}). \quad (3)$$

Introducing a nonnegative multiplier for the constraint gives the regularized objective. For $\lambda > 0$, we write

$$\pi_\lambda \in \arg \min_{\pi \in \Pi(\mathbf{P}^{\text{blk}}, \mathbf{u}_m)} \left\{ \sum_{b=1}^B \sum_{j=1}^m \pi(b, j) c(b, j) + \lambda D_{\text{KL}}(\pi \| \mathbf{P}^{\text{blk}} \otimes \mathbf{u}_m) \right\}. \quad (4)$$

Here β bounds the allowed entropy loss, while λ penalizes dependence in the objective. Increasing β permits more entropy loss; increasing λ cannot increase the loss at an exact optimum (Proposition A.1). A zero budget gives the independent coupling. Positive budgets need not be attained exactly, because the largest attainable loss depends on the block and column marginals.

The numerical solver alternates coupling updates with adjustments to λ . It restores the prescribed marginals before sampling and reports the achieved entropy loss. A finite iteration limit or numerical tolerance can leave that loss above or below the target, so the requested budget alone is not a guarantee for the numerical output. Appendix A.1 describes the procedure, and Theorem A.3 gives a conditional convergence bound.

Token sampling. The key selects a column J that is uniform conditional on the partition and cost table in the idealized construction. For any coupling π with the prescribed marginals, the conditional block probability is $r(b \mid j) = m\pi(b, j)$. Sampling within the chosen block according to the original relative probabilities gives

$$Q(w \mid j) = m\pi\{g(w), j\} \frac{P_w}{P_{g(w)}^{\text{blk}}}, \quad w \in \mathcal{A}, \quad (5)$$

with $Q(w \mid j) = 0$ outside $\mathcal{A}$. The denominator is positive for $w \in \mathcal{A}$. This formula applies to both exact OT solutions and numerically computed couplings after marginal correction.

For a fixed partition and cost table, averaging over J recovers the NTP distribution: $m^{-1} \sum_j Q(w \mid j) = P_w$. Moreover,

$$\frac{1}{m} \sum_{j=1}^m H\{Q(\cdot \mid j)\} = H(P) - D_{\text{KL}}(\pi \parallel P^{\text{blk}} \otimes \mathbf{u}_m).$$

A coupling satisfying (3) thus retains at least a fraction $1 - \beta$ of NTP entropy on average across columns. This is not a lower bound for every column. Preservation of the entire sequence distribution additionally requires conditional uniformity at each generation step, as stated in Theorem A.4.

Compatibility with speculative sampling. textGrain can be combined with speculative sampling while the draft and target models share a secret key (Leviathan et al., 2023; Chen et al., 2023). The draft proposes tokens, and the target evaluates the proposals in parallel. Verification uses the two watermarked conditional distributions to preserve the target’s watermarked output law. Appendix B gives the algorithm and its assumptions. The resulting speed depends on proposal acceptance and the cost of watermarking both models; distributional equivalence alone does not establish a speedup.

3.2 Watermark detection

Reconstructing the scores. The coupling favors pairs of blocks and columns with smaller costs, so detection tests whether observed tokens tend to occupy such pairs (Figure 2). The textGrain detector uses the same tokenizer, block and column counts, context-window rule, and keyed pseudo-random construction as generation. At each scored position, the key and observed context window reconstruct the partition, cost table, and selected column. The token identifies its block and hence the relevant cost. Reconstruction requires neither the coupling nor the generating model or entropy budget.

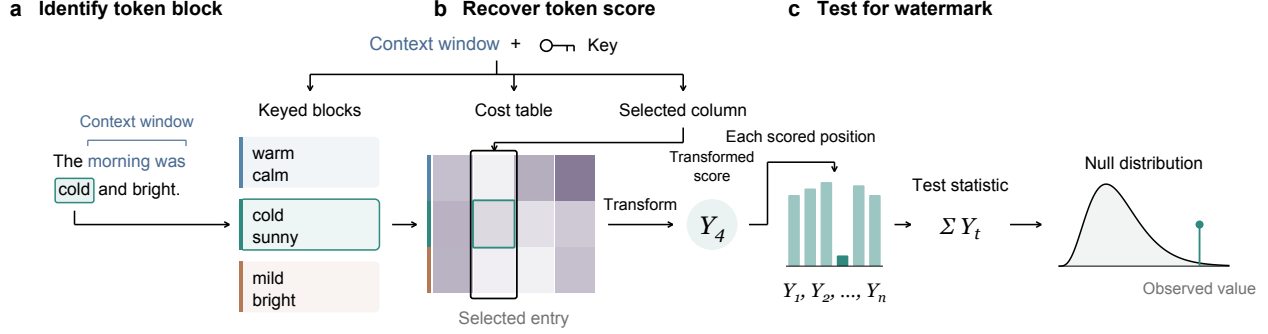


Figure 2: **Detecting the watermark from the observed text.** **a**, The detector identifies the block containing the observed token. As in Figure 1, blue text marks a context window of two tokens. **b**, The context window and key reconstruct the cost table and selected column (black outline). The cost for the observed block (jade outline) is transformed into the score Y_4 for the fourth illustrated token, cold. **c**, Repeating this procedure at the scored positions yields $Y_1, \dots, Y_n$. Their sum is compared with its distribution under unwatermarked text to test for a watermark.

Detection statistic and calibration. Let c_t be the reconstructed cost at scored position t . Undoing standardization gives the original Gumbel value $Z_t = \gamma - (\pi/\sqrt{6})c_t$. We define

$$Y_t = -\log\{1 - F_G(Z_t)\}, \quad S_n = \sum_{t=1}^n Y_t,$$

where n is the number of scored positions. Smaller costs yield larger scores and therefore larger values of S_n .

For the idealized null calculation, assume the unwatermarked text is independent of the secret key and that the independence conditions for the partition, cost table, and column hold. Conditional on the text, each reconstructed Z_t is then a standard Gumbel variable, so Y_t has the unit exponential distribution. To avoid reusing the same table, score only the first occurrence of each distinct context window. Under the assumed independence across distinct contexts, the resulting scores are independent and $S_n \sim \text{Gamma}(n, 1)$, with shape n and scale 1. For $n \geq 1$, we declare a watermark when S_n exceeds this distribution’s $(1 - \alpha)$ quantile. The conditional false positive rate is α under these assumptions; with no scored positions, the test makes no detection.

Ignoring repeated contexts during detection does not enforce the conditional uniformity needed for sequence preservation in Theorem A.4. One way to obtain that property is to sample directly from the NTP distribution at contexts already used during generation. A fixed deployed key and finite precision arithmetic require empirical calibration checks; the idealized null calculation does not by itself guarantee the same error rate for every key or application.

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A Block OT with entropy calibration: solver and guarantees

A.1 Solving the OT problem under an entropy constraint

We solve the OT problem with KL regularization in (4) while adjusting λ towards the target entropy loss. The method alternates between updating the coupling and adjusting λ according to the achieved loss. Throughout this appendix, $\boldsymbol{\rho} = \boldsymbol{P}^{\text{blk}}$ denotes the block distribution, $\boldsymbol{u}_m$ the uniform column distribution, and $c(b, j) = -G_{bj}$. For $\lambda > 0$ and a finite cost table, the equivalent maximization problem is

$$\boldsymbol{\pi}_\lambda \in \arg \max_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\rho}, \boldsymbol{u}_m)} \{ \langle \boldsymbol{\pi}, \boldsymbol{G} \rangle - \lambda D_{\text{KL}}(\boldsymbol{\pi} \| \boldsymbol{\rho} \otimes \boldsymbol{u}_m) \}. \quad (\text{A.1})$$

Algorithm A.1 gives the procedure for one token position.

Initialization and Sinkhorn updates. Line 1 excludes blocks with zero mass before the iterations. A zero target or zero block entropy gives the independent coupling. In particular, the algorithm returns before dividing by $H(\boldsymbol{P})$ when this entropy is zero. Otherwise, line 5 caps the requested loss at $\min\{H(\boldsymbol{\rho}), \log m\}$ and, relative to block entropy, at $1 - 10^{-8}$. These caps follow from the information bound in Theorem A.4. They can reduce the requested target, but do not guarantee that the remaining target is attainable for the given marginals, cost table and interval of λ .

For fixed $\lambda > 0$, the matrix $K_\lambda(b, j) = \exp\{-c(b, j)/\lambda\}$ gives the solution the form $\boldsymbol{\pi}_\lambda = \text{diag}(\boldsymbol{u}) \boldsymbol{K}_\lambda \text{diag}(\boldsymbol{v})$. The row and column scaling vectors $\boldsymbol{u}, \boldsymbol{v}$ enforce the prescribed marginals through alternating Sinkhorn updates (Cuturi, 2013). Lines 11–14 perform these updates and normalize each column to mass $1/m$. Evaluating them in the log domain using log-sum-exp avoids forming very large or small kernel entries explicitly. Any probability floor used when taking logarithms is a numerical modification of the update. Line 12 subtracts the mean of $\log \boldsymbol{v}$. The subsequent row update cancels this rescaling, leaving the coupling unchanged in exact arithmetic.

Marginal correction. After these updates, a finite iterate may still have incorrect row sums. Line 15 restores both prescribed marginals by scaling down rows and columns with excess mass, then redistributing the missing mass to fill both sets of deficits. We denote this operation by $\mathcal{R}$ and its input table by $\boldsymbol{X}$.

Algorithm A.1 Alternating updates of the block coupling and regularization

Require: Block probabilities $\mathbf{P}^{\text{blk}}$, cost table c , NTP entropy $H(\mathbf{P})$ and target $\beta \in [0, 1]$

```

1: Remove blocks with zero mass and the corresponding rows of  $c$ 
2: if  $\beta = 0$  or  $H(\mathbf{P}^{\text{blk}}) = 0$  then
3:   return  $\mathbf{P}^{\text{blk}} \otimes \mathbf{u}_m$ , zero entropy loss and status “independent coupling”
4: end if
5:  $\beta \leftarrow \min\{\beta, (1 - 10^{-8})H(\mathbf{P}^{\text{blk}})/H(\mathbf{P}), \log m/H(\mathbf{P})\}$ 
6: If  $\beta = 0$ , return the independent coupling with zero entropy loss
7: Set  $\lambda = 1$ ,  $\mathbf{v} = \mathbf{u}_m$ ,  $a = 1/2$ ,  $\lambda_{\min} = 10^{-2}$ ,  $\lambda_{\max} = 10^2$ 
8: Set  $\delta = 10^{-12}H(\mathbf{P}^{\text{blk}})/H(\mathbf{P})$ , tolerance  $\varepsilon_I$  and iteration limit  $N_{\text{it}} \geq 1$ 
9: for  $s = 0, \dots, N_{\text{it}} - 1$  do
10:   Form  $\log \mathbf{K}_\lambda = [-c(b, j)/\lambda]_{bj}$  ▷ Evaluate updates in the log domain
11:    $\mathbf{u} \leftarrow \mathbf{P}^{\text{blk}}/(\mathbf{K}_\lambda \mathbf{v})$ ;  $\mathbf{v} \leftarrow \mathbf{u}_m/(\mathbf{K}_\lambda^\top \mathbf{u})$  ▷ Update row, then column factors
12:    $\log \mathbf{v} \leftarrow \log \mathbf{v} - m^{-1}(\mathbf{1}^\top \log \mathbf{v})\mathbf{1}$  ▷ Center the column factors
13:    $\mathbf{u} \leftarrow \mathbf{P}^{\text{blk}}/(\mathbf{K}_\lambda \mathbf{v})$  ▷ Refresh row factors
14:   Rescale each column of  $\text{diag}(\mathbf{u})\mathbf{K}_\lambda$  independently to sum to  $1/m$ , giving  $\boldsymbol{\pi}_s$ 
15:    $\tilde{\boldsymbol{\pi}}_s \leftarrow \mathcal{R}(\boldsymbol{\pi}_s)$  using (A.2) ▷ Restore both marginals
16:   Compute the fraction of entropy removed,  $\hat{\beta}_s$ , using (A.3)
17:   if  $|\hat{\beta}_s - \beta| \leq \varepsilon_I$  or  $s = N_{\text{it}} - 1$  then
18:     return  $\tilde{\boldsymbol{\pi}}_s$ ,  $\hat{\beta}_s$  and whether  $|\hat{\beta}_s - \beta| \leq \varepsilon_I$ 
19:   end if
20:    $\lambda_{\text{new}} \leftarrow \text{clip}_{[\lambda_{\min}, \lambda_{\max}]}[\lambda \{\max(\hat{\beta}_s, \delta)/\beta\}^a]$  ▷ Adjust entropy loss
21:   Rescale  $\mathbf{v} \leftarrow \exp\{(\lambda/\lambda_{\text{new}}) \log \mathbf{v}\}$ ; then set  $\lambda \leftarrow \lambda_{\text{new}}$ 
22: end for

```

Let $\mathbf{X} \geq 0$ have total mass one, with prescribed row and column marginals $\mathbf{a} = \boldsymbol{\rho}$ and $\mathbf{b} = \mathbf{u}_m$. The vectors $\mathbf{x}$ and $\mathbf{y}$ below contain the row and column scaling factors. Each factor is at most one, so these steps only remove excess mass:

$$\begin{aligned}
\mathbf{X}' &= \text{diag}(\mathbf{x})\mathbf{X}, & x_i &= \min\left\{\frac{a_i}{(\mathbf{X}\mathbf{1})_i}, 1\right\}, \\
\mathbf{X}'' &= \mathbf{X}' \text{diag}(\mathbf{y}), & y_j &= \min\left\{\frac{b_j}{((\mathbf{X}')^\top \mathbf{1})_j}, 1\right\}.
\end{aligned}$$

If a denominator is zero, set the corresponding scale to one. The remaining marginal deficits $\mathbf{e}_a = \mathbf{a} - \mathbf{X}''\mathbf{1}$ and $\mathbf{e}_b = \mathbf{b} - (\mathbf{X}'')^\top \mathbf{1}$ are nonnegative and have the same total mass $\Delta = \mathbf{1}^\top \mathbf{e}_a = \mathbf{1}^\top \mathbf{e}_b$. Distributing this missing mass according to the two deficit vectors gives the corrected coupling

$$\mathcal{R}(\mathbf{X}) = \begin{cases} \mathbf{X}'' + \mathbf{e}_a \mathbf{e}_b^\top / \Delta, & \Delta > 0, \\ \mathbf{X}'', & \Delta = 0. \end{cases} \quad (\text{A.2})$$

It is nonnegative and satisfies $\mathcal{R}(\mathbf{X})\mathbf{1} = \mathbf{a}$ and $\mathcal{R}(\mathbf{X})^\top \mathbf{1} = \mathbf{b}$. Moreover,

$$\|\mathcal{R}(\mathbf{X}) - \mathbf{X}\|_1 \leq 2(\|\mathbf{X}\mathbf{1} - \mathbf{a}\|_1 + \|\mathbf{X}^\top \mathbf{1} - \mathbf{b}\|_1),$$

because scaling down the rows and columns removes at most their excess mass, and the rank-one update fills the remaining deficit (Altschuler et al., 2017, Algorithm 2 and Lemma 7). Thus vanishing marginal residuals imply a vanishing correction. Convergence to the OT optimum additionally

requires convergence of the underlying iterates. At iteration s , applying this correction to the current transport table gives the coupling $\tilde{\pi}_s$ used in Algorithm A.1. Entropy loss is evaluated on this corrected table. Both marginal residuals should also be checked in floating point arithmetic.

Entropy loss and regularization updates. Line 16 measures the fraction of entropy removed by the corrected coupling,

$$\hat{\beta}_s = \frac{D_{\text{KL}}(\tilde{\pi}_s \| \mathbf{P}^{\text{blk}} \otimes \mathbf{u}_m)}{H(\mathbf{P})}. \quad (\text{A.3})$$

The same calculation can be performed by dividing the KL divergence by block entropy $H(\boldsymbol{\rho})$ and rescaling the target and tolerance by $H(\mathbf{P})/H(\boldsymbol{\rho})$. Rescaling all three quantities together preserves both the stopping test and the ratio of achieved loss to target loss used to update λ . The reported fraction uses the NTP entropy normalization above.

Line 20 increases λ when the loss exceeds the target and decreases it when the loss is too small. The exponent $a = 1/2$ moderates the adjustment, while the floor δ and clipping bounds keep it well defined. When normalizing by block entropy, the floor is 10^{-12} , equivalent to $\delta = 10^{-12}H(\boldsymbol{\rho})/H(\mathbf{P})$ in the algorithm. Line 21 multiplies $\log \mathbf{v}$ by $\lambda_{\text{old}}/\lambda_{\text{new}}$, preserving $\lambda \log \mathbf{v}$ so that the next Sinkhorn step reuses the current factors. The algorithm applies marginal correction before sampling and returns the corrected coupling, achieved loss and convergence status. Active numerical floors and bounds can affect its output.

Marginal correction enforces the prescribed marginals, but does not enforce the entropy budget. Even when the stopping test succeeds, the achieved fraction can be as large as $\beta + \varepsilon_I$; after N_{it} iterations, it can differ by more. The entropy identity therefore applies with the achieved fraction $\hat{\beta}_s$. A strict budget claim requires checking $\hat{\beta}_s \leq \beta$ separately. Both the original request and any capped target should be reported with the achieved loss and convergence status.

Batching and computational cost. Once several NTP distributions are available, their OT problems can be solved in parallel by stacking the cost tables and scaling vectors. Each problem keeps its own λ , target entropy loss and stopping criterion. Problems that reach tolerance stop updating while the others continue. Tables with different numbers of blocks are padded, and padded rows are excluded from the updates. This combines independent solves into shared matrix operations without changing their results, apart from floating point error. It does not remove the sequential dependence needed to obtain NTP distributions during ordinary autoregressive generation.

Each Sinkhorn update and marginal correction requires $O(Bm)$ operations. Aggregating the active vocabulary and performing N_{it} updates therefore costs $O(|\mathcal{A}| + N_{\text{it}}Bm)$, where $\mathcal{A}$ is the active support.

A.2 Calibrating entropy loss

We adjust λ to bring the achieved entropy loss towards its target. The KL divergence of the exact coupling from independence equals the mutual information between block and column. It also equals the average token entropy lost across columns. For $H(\mathbf{P}) > 0$, write $\theta = \log \lambda$ and let $\beta_{\text{eff}}(\theta) = D_{\text{KL}}(\pi_{e^\theta} \| \boldsymbol{\rho} \otimes \mathbf{u}_m) / H(\mathbf{P})$ be the fraction of entropy removed by the exact OT solution. A finite number of Sinkhorn steps and marginal correction give the approximation $\hat{\beta}_s$. For a positive target, the update in line 20 is then

$$\theta_{s+1} = \text{proj}_\Theta \left\{ \theta_s + a \log \left(\frac{\hat{\beta}_s}{\beta} \right) \right\}, \quad (\text{A.4})$$

where a is the step size and $\Theta = [\log \lambda_{\min}, \log \lambda_{\max}]$ is the permitted interval for $\log \lambda$. This expression applies when the numerical floor is inactive, with β denoting the target after any cap. Normalizing by block or token entropy gives the same ratio in the update. The solver uses matrix scaling (Cuturi, 2013), stabilization in the log domain (Schmitzer, 2019) and marginal correction (Altschuler et al., 2017), together with the update in (A.4). Rescaling the stored factors lets successive updates reuse the current solution. The direction of the update follows from a simple property: increasing λ cannot increase the KL divergence of the optimal coupling from independence.

Proposition A.1 (Monotonicity of the KL divergence). *For a fixed finite cost table and marginals restricted to their positive support, (A.1) has a unique optimizer for each $\lambda > 0$. Its KL divergence $D_{\text{KL}}(\pi_\lambda \| \rho \otimes \mathbf{u}_m)$ is nonincreasing in λ . It is strictly decreasing between any two values of λ whose optimal couplings differ. Hence a target KL divergence determines at most one λ on every strictly decreasing portion of the path.*

Proof of Proposition A.1. The feasible set is compact, and the objective is continuous and strictly concave for $\lambda > 0$, giving existence and uniqueness. For $0 < \lambda_1 < \lambda_2$, write $\pi_i = \pi_{\lambda_i}$, $S_i = \langle \pi_i, \mathbf{G} \rangle$, and $D_i = D_{\text{KL}}(\pi_i \| \rho \otimes \mathbf{u}_m)$. Optimality gives $S_1 - \lambda_1 D_1 \geq S_2 - \lambda_1 D_2$ and $S_2 - \lambda_2 D_2 \geq S_1 - \lambda_2 D_1$. Adding yields $(\lambda_2 - \lambda_1)(D_1 - D_2) \geq 0$, hence $D_1 \geq D_2$. If $D_1 = D_2$, the two inequalities force $S_1 = S_2$, so π_2 is also optimal at λ_1 . Uniqueness then gives $\pi_1 = \pi_2$. Hence $D_1 > D_2$ whenever the optimal couplings differ. $\square$

The update increases λ when entropy loss exceeds the target and decreases it when the loss is too small. Its convergence rate depends on how strongly the loss responds to changes in λ . We measure this response by $\nu(\theta) := -d \log \beta_{\text{eff}}(\theta) / d\theta$. For example, $\nu \approx 2$ means that a small 1% increase in λ reduces the fraction of entropy removed by about 2%. The following condition rules out a response that is too flat or too steep for the chosen step size.

Assumption A.2 (Regularity of the response to regularization). The exact fraction $\beta_{\text{eff}}(\theta)$ is positive and continuously differentiable on a closed interval Θ containing the initial iterate θ_0 and a solution θ^* of $\beta_{\text{eff}}(\theta^*) = \beta$. Its relative sensitivity $\nu(\theta)$ and the step size $a > 0$ satisfy

$$0 < \underline{\nu} \leq \nu(\theta) \leq \bar{\nu} < \infty \quad (\theta \in \Theta), \quad q := \max\{|1 - a\underline{\nu}|, |1 - a\bar{\nu}|\} < 1.$$

The estimate after marginal correction is positive and satisfies $\log \hat{\beta}_s = \log \beta_{\text{eff}}(\theta_s) + e_s$, where e_s is the error caused by the approximate OT solution, measured on the logarithmic scale.

This assumption requires an attainable target and quantitative bounds on the response throughout the chosen interval. Monotonicity alone does not imply these bounds. We use $a = 1/2$ to damp the update; the following convergence guarantee applies only when the stated sensitivity and approximation conditions hold.

Theorem A.3 (Convergence of the regularization update). *Under Assumption A.2, the iterates in (A.4) satisfy*

$$|\theta_s - \theta^*| \leq q^s |\theta_0 - \theta^*| + a \sum_{j=0}^{s-1} q^{s-1-j} |e_j|.$$

Thus exact OT solutions ($e_s = 0$) give geometric convergence with contraction factor q . Uniformly bounded approximation errors limit the guaranteed accuracy, while errors $|e_j| = O(r^j)$, $0 < r < 1$, contribute $O(\max\{q, r\}^s)$ when $r \neq q$ and $O(sq^{s-1})$ when $r = q$.

Proof of Theorem A.3. Put $\phi(\theta) = \log \beta_{\text{eff}}(\theta) - \log \beta$. Then $\phi(\theta^*) = 0$ and $\phi'(\theta) = -\nu(\theta)$. Before projection, the mean value theorem gives $\theta_s + a\phi(\theta_s) - \theta^* = \{1 - a\nu(\xi_s)\}(\theta_s - \theta^*)$ for some ξ_s between θ_s and θ^* . Projection onto an interval containing θ^* is nonexpansive, so Assumption A.2 gives a contraction at each step with the additional perturbation ae_s . Iterating this bound proves the result. $\square$

This bound separates two sources of error: the initial choice of λ and the inaccuracy of each OT solution. The first decays geometrically. The second accumulates with decreasing weights, so accurate calibration also requires sufficiently accurate Sinkhorn iterates. The theorem does not establish convergence of a fixed number of alternating updates without control of the errors e_s .

Near the largest attainable entropy loss, changing λ may have little effect on the coupling. Then ν can approach zero and convergence can slow. The achieved loss should therefore be reported alongside the requested target. At a fixed λ , a large cost range relative to λ makes the entries of $\mathbf{K}_\lambda = \exp(\mathbf{G}/\lambda)$ more uneven and can slow Sinkhorn convergence (Peyré and Cuturi, 2019). Reusing the previous scaling factors provides a nearby starting point when successive values of λ are close and the solution varies smoothly.

A.3 Marginal preservation and entropy loss

Sampling within each block according to the original token probabilities preserves the token marginal of the corrected coupling. It also leaves the block choice as the only source of dependence between the token and the selected column. The following result makes these statements precise and identifies the resulting loss of conditional entropy.

Theorem A.4 (Marginal preservation and the entropy identity). *Fix an NTP distribution $\mathbf{P}$ with support $\mathcal{A}$, a partition g with block law $\boldsymbol{\rho}$, and conditional block laws $\mathbf{r}(\cdot | j)$ satisfying $m^{-1} \sum_{j=1}^m r(b | j) = \rho_b$. Let J be uniform on $[m]$, set $\pi(b, j) = r(b | j)/m$, and sample W from*

$$Q(w | j) = r\{g(w) | j\} \frac{P_w}{\rho_{g(w)}}, \quad w \in \mathcal{A},$$

with $Q(w | j) = 0$ outside $\mathcal{A}$. The denominator is positive on $\mathcal{A}$. Writing $C = g(W)$, we have

$$\begin{aligned} \frac{1}{m} \sum_{j=1}^m Q(w | j) &= P_w, \\ I(W; J) &= I(C; J) = D_{\text{KL}}(\boldsymbol{\pi} \| \boldsymbol{\rho} \otimes \mathbf{u}_m) \\ &= H(\mathbf{P}) - \mathbb{E}_J H\{\mathbf{Q}(\cdot | J)\} \\ &\leq \min\{H(\boldsymbol{\rho}), \log m\}. \end{aligned}$$

Consequently, for $H(\mathbf{P}) > 0$, an entropy loss of $I(W; J) = \beta_{\text{eff}} H(\mathbf{P})$ retains the fraction $1 - \beta_{\text{eff}}$ of NTP entropy on average over columns. This is not a bound for each individual column or key. If, at every unmasked position, the column index is uniform conditionally on the preceding generation history and on the current partition and transport table, and if masked positions sample directly from the NTP distribution, then the generated sequence, averaged over keys, has the same law as unwatermarked generation.

Proof of Theorem A.4. For $w \in \mathcal{A}$ and $b = g(w)$, substituting the lift and using the block marginal gives $m^{-1} \sum_j Q(w | j) = P_w$; both sides vanish outside $\mathcal{A}$. The joint law factors as $\Pr(J) \Pr(C | J) P(W | C)$, so $W \perp J | C$. Since C is a function of W , $I(W; J) = I(C; J)$. The law of (C, J) is $\boldsymbol{\pi}$

with marginals ρ and $\mathbf{u}_m$, giving the KL identity. The entropy identity follows from the preserved marginal $W \sim \mathbf{P}$, and the bound follows from $I(C; J) \leq \min\{H(C), H(J)\} = \min\{H(\rho), \log m\}$. Conditioning first on the generation history, partition and table permits the same averaging over columns at each unmasked position. Masked positions follow the NTP distribution by assumption. Iterating these conditional identities proves the sequence statement. $\square$

The sequence statement requires conditional uniformity after observing the generated history. Reusing the pseudorandom values for a repeated context can violate this condition, even though the identity at a fixed prefix remains valid. Fresh independent values at each position, or masking repeated contexts under an ideal model with independent randomness for distinct contexts, provide sufficient conditions. A practical pseudorandom function approximates this ideal model; the theorem does not assert exact independence for a fixed key.

B Speculative sampling with a shared key

B.1 Generation and block verification

We combine the block watermark with speculative sampling by watermarking both the draft and target distributions with the same secret key k . Both models apply marginal correction before sampling or evaluating token probabilities. Their NTP distributions and target entropy losses may differ.

Let $\mathbf{P}_i^d$ and $\mathbf{P}_i^t$ be the draft and target NTP distributions at the initial prefix followed by the first $i - 1$ proposals. Denote their watermarked distributions conditional on k by $\mathbf{Q}_i^d$ and $\mathbf{Q}_i^t$. Both models use the same key and construction of pseudorandom values. Their different NTP distributions can nevertheless yield different conditional token distributions. Verification uses the watermarked probabilities that govern sampling, as required by Proposition B.1.

Algorithm B.1 Speculative sampling with a shared watermark key

Require: Draft and target models, shared key k , draft length $K \geq 1$, and both watermark configurations

- 1: **for** $i = 1, \dots, K$ **do**
 - 2: Compute $\mathbf{P}_i^d$ and its corrected watermarked distribution $\mathbf{Q}_i^d$ using k .
 - 3: Draw $x_i \sim \mathbf{Q}_i^d$ with fresh sampling randomness.
 - 4: **end for**
 - 5: Compute $\mathbf{P}_1^t, \dots, \mathbf{P}_{K+1}^t$ in parallel in one target forward pass.
 - 6: Apply the watermark with key k to obtain $\mathbf{Q}_1^t, \dots, \mathbf{Q}_{K+1}^t$.
 - 7: Compute r_i, h_i by (B.1)–(B.2).
 - 8: Draw independent $U_i \sim \text{Uniform}(0, 1)$ and set $L_{\text{keep}} = \max(\{0\} \cup \{i : U_i \leq h_i\})$.
 - 9: **if** $L_{\text{keep}} < K$ **then**
 - 10: Draw $W \sim \mathbf{R}_{L_{\text{keep}}}$ from (B.3).
 - 11: **else**
 - 12: Draw $W \sim \mathbf{Q}_{K+1}^t$. $\triangleright$ Bonus token
 - 13: **end if**
 - 14: Emit $(x_1, \dots, x_{L_{\text{keep}}}, W)$ and discard the remaining proposals.
-

Drafting and target evaluation. The draft proposes K tokens sequentially from its watermarked distributions. The target then evaluates all draft positions and the possible bonus position

in one forward pass. Once these NTP distributions are available, their OT problems can also be solved in a batch. Proposal draws, verification uniforms and the final draw use fresh independent sampling randomness. The models must use compatible token identifiers and input formatting, and probabilities must reflect the temperature, truncation and watermarking used for generation.

Joint verification. Block verification tests candidate prefix lengths jointly and can retain a longer prefix even if a shorter one fails its test (Sun et al., 2025). Starting from $r_0 = 1$, define

$$r_i = \min \left\{ r_{i-1} \frac{Q_i^t(x_i)}{Q_i^d(x_i)}, 1 \right\}, \quad i = 1, \dots, K. \quad (\text{B.1})$$

The denominator is positive for every proposal that can be sampled. The quantity r_i can decrease for a token less likely under the target and recover when later proposals favor it. For $1 \leq i < K$, set

$$z_i = \sum_{w \in \mathcal{V}} [r_i Q_{i+1}^t(w) - Q_{i+1}^d(w)]_+, \quad h_i = \frac{z_i}{z_i + 1 - r_i}, \quad h_K = r_K, \quad (\text{B.2})$$

where $[a]_+ = \max(a, 0)$ and $h_i = 1$ if its numerator and denominator both vanish. Independent tests $U_i \leq h_i$ determine the longest retained prefix L_{keep} . A failed test at one length does not rule out retaining a longer prefix. These tests and the following residual must be used together to preserve the target law.

Correction and bonus tokens. If $L_{\text{keep}} < K$, the remaining proposals are discarded and the next token is drawn from

$$R_{L_{\text{keep}}}(w) = \frac{[r_{L_{\text{keep}}} Q_{L_{\text{keep}}+1}^t(w) - Q_{L_{\text{keep}}+1}^d(w)]_+}{\sum_{v \in \mathcal{V}} [r_{L_{\text{keep}}} Q_{L_{\text{keep}}+1}^t(v) - Q_{L_{\text{keep}}+1}^d(v)]_+}. \quad (\text{B.3})$$

This residual supplies the target mass missing from the draft. Its denominator is positive whenever this branch has positive probability. No additional watermark transformation is applied to the residual, because it is already derived from the watermarked distributions. If all K proposals are retained, a bonus token is drawn directly from Q_{K+1}^t without another target forward pass. When caching is enabled, the model caches are restored to the retained prefix before continuing, and stopping rules are applied to emitted tokens in order.

B.2 Distribution preservation and detection

Proposition B.1 (Preservation of the watermarked target law). *Fix a key k . Suppose each proposal is sampled from the conditional distribution Q_i^d used in verification, each target distribution Q_i^t is the one used by direct watermarked target generation at the same prefix, and all draws use fresh independent sampling randomness. With exact probability calculations, repeated application of Algorithm B.1 generates the same law for the output sequence as direct sampling from the target’s watermarked distributions conditional on k .*

Proof of Proposition B.1. Fix k and suppress it from the notation. Let $X_1, \dots, X_K$ denote the draft proposals. We first verify the retention identity from Sun et al. (2025, Lemma 3),

$$\Pr(L_{\text{keep}} \geq i \mid X_{1:i} = x_{1:i}) = r_i,$$

where later proposals and verification uniforms are averaged out. The identity holds for $i = K$ because $h_K = r_K$. For the induction step, the recursion in (B.1) gives

$$Q_{i+1}^d(w)r_{i+1}(x_{1:i}, w) = \min\{Q_{i+1}^d(w), r_i Q_{i+1}^t(w)\}.$$

Assuming the identity at $i + 1$, summing over w gives

$$\Pr(L_{\text{keep}} > i \mid X_{1:i} = x_{1:i}) = r_i - z_i.$$

The uniform U_i is independent of all later proposals and tests. The probability of retaining at least i proposals is therefore

$$(r_i - z_i) + (1 - r_i + z_i)h_i = r_i,$$

including the case $r_i = 1, z_i = 0$. Backward induction proves the identity. Hence the probability of correction at length i , conditional on the proposed prefix, is z_i . For $i = 0$, define z_0 by the same positive-part sum as in (B.2), with $r_0 = 1$; the same conclusion follows from the identity at $i = 1$.

The residual fills the missing mass because, for $0 \leq i < K$,

$$\underbrace{Q_{i+1}^d(w)r_{i+1}(x_{1:i}, w)}_{\text{retained proposal}} + \underbrace{z_i R_i(w)}_{\text{correction}} = r_i Q_{i+1}^t(w).$$

For tokens with $Q_{i+1}^d(w) = 0$, the retained proposal contributes zero and the residual supplies the required target mass. When $z_i = 0$, correction at that prefix has probability zero and its contribution is defined as zero. Starting from $r_0 = 1$, this balance gives the target conditional distribution for the next output token. After a correction, the procedure restarts at the emitted prefix; after K retained proposals, the bonus token follows the target law. Induction over emitted tokens establishes the sequence law (Sun et al., 2025, Theorem 1). The argument holds conditional on the shared key and does not require separate draft and target keys. $\square$

Thus different draft settings affect acceptance and computational cost while preserving the ideal target output law. The OT solves themselves need not reach the exact optimum: the proposition applies to the corrected distributions actually used, provided that sampling and verification use the same probabilities and that the target distribution matches direct target generation at each prefix. Numerical errors in evaluating acceptance probabilities or normalizing residuals can affect the identity. Averaging over keys recovers the same distribution as direct watermarked target generation. Equality with unwatermarked generation additionally requires the conditional uniformity assumptions in Theorem A.4.

Detection uses the shared key and the target watermark configuration, following Section 3.2. It does not require draft tokens to be identified or iteration boundaries to be recorded. Under the proposition’s conditions, the same detector calibration and definitions of diversity apply as for direct watermarked target generation. Retained entropy is determined by the target’s achieved entropy loss. Acceptance rate and elapsed time require empirical measurement.

C Related work

C.1 Language model watermarks

Early text watermarks modified completed passages through syntactic transformations or synonym substitution (Atallah et al., 2001; Topkara et al., 2006). Modern generative watermarks intervene during decoding. At each step, a pseudorandom function of the secret key and context generates

reproducible values that guide token sampling. The detector reconstructs these values from the text and key and tests for dependence between them and the observed tokens (Li et al., 2025a).

Different methods address different objectives. Green list methods bias generation toward a keyed subset of tokens to improve detectability while maintaining text quality (Kirchenbauer et al., 2023, 2024). Adaptive methods and variants that assign different strengths to individual tokens adjust this bias across contexts (Liu and Bu, 2024; Huo et al., 2024). Watermarks that carry several bits can encode identifiers or metadata for attribution, rather than only indicating the presence of a watermark (Fernandez et al., 2023; Yoo et al., 2023; Xu et al., 2026; Chen et al., 2026). Cryptographic approaches study watermarks that are computationally undetectable without the key, resistance to adversarial changes, and limits on secure watermarking (Christ et al., 2024; Christ and Gunn, 2024; Zhang et al., 2024). Semantic methods use features or sentence representations that can be less sensitive to changes in individual tokens, with the aim of retaining a signal after edits such as paraphrasing (Liu et al., 2024; Hou et al., 2024; Dabiriaghdam and Wang, 2025; Nemecek et al., 2026).

We study how watermarking changes the sampling diversity available under a fixed key. Our construction embeds a binary provenance signal during generation and preserves the NTP distribution after averaging over keys. Its entropy budget measures the average sampling randomness removed to create dependence on the key.

C.2 Unbiasedness and diversity under a fixed key

Green list methods generally change the NTP distribution even after averaging over their keyed token subsets (Kirchenbauer et al., 2023, 2024). We focus on methods that preserve this distribution after averaging over keyed randomness. At a fixed prefix, the randomly drawn key induces $\Xi \sim \mu$, as in Section 2. Such methods are variously called unbiased, distribution preserving, or distortion free; we use unbiased, following the statistical literature (Hu et al., 2024; Li et al., 2025a). An unbiased watermark defines a coupling of Ξ and the generated token W whose token marginal is the original NTP distribution. Examples include Gumbel-max sampling (Aaronson, 2023; Sander et al., 2026), reweighting schemes (Hu et al., 2024; Wu et al., 2024; Chen et al., 2025), inverse transform sampling (Kuditipudi et al., 2024), optimized couplings (Xie et al., 2025; Tsur et al., 2025), and tournament sampling (Dathathri et al., 2024). MCmark reweights token probabilities by favoring a vocabulary segment selected using the key, with the adjustments designed to preserve the NTP distribution on average (Chen et al., 2025). Our method also groups tokens, but determines the conditional block probabilities through an OT problem whose marginal constraints preserve the NTP distribution and whose entropy-loss budget controls the sampling randomness retained under a fixed key.

Unbiasedness at each fixed prefix does not by itself establish preservation of the entire sequence distribution. Repeated contexts can reuse the same keyed values, so their conditional distribution depends on the preceding generation history (Cai et al., 2025). Dathathri et al. (2024, Supplementary Appendices G.2 and G.3) distinguish these guarantees and mask repeated context windows to preserve the sequence distribution. Both guarantees average over keyed randomness. Neither directly controls the entropy $H\{Q(\cdot | \xi)\}$ that remains after the prefix and keyed values are fixed. We quantify its reduction from $H(\mathbf{P})$ on average over Ξ .

Fu et al. (2024) address the lack of diversity in Gumbel-max generation under a fixed key. GumbelSoft samples from a softmax distribution over perturbed logits instead of taking the hard maximum. This restores sampling randomness but does not generally preserve the original NTP distribution, which separates its objective from unbiased entropy control.

Recent work also optimizes the coupling directly. He et al. (2025a) jointly optimize the watermarking scheme and detector under a distortion constraint, obtaining an unbiased construction as

the zero distortion case. Long et al. (2025) optimize the worst case detection power over a class of NTP distributions. Huang et al. (2026) jointly optimize a coupling and an e-value for detection that remains valid under optional stopping. Tsur et al. (2025) use OT with costs based on coding theory or distributions with heavy tails. Gloaguen et al. (2026) consider a broader constrained optimization framework that allows both unbiased and biased watermarks. Their proposed PPL-based methods do not generally preserve the original NTP distribution, because controlling expected log-likelihood does not ensure unbiasedness. We require this distribution to be preserved and use OT to control the sampling entropy retained under a fixed key.

Our construction uses the coupling’s KL divergence from independence to control the average entropy loss. Applying this constraint to blocks of tokens reduces the optimization dimension while preserving relative probabilities within each block.

C.3 Statistical inference for language model watermarks

Watermark detection tests for dependence between text and keyed randomness, often through a statistic whose null distribution does not depend on the unknown NTP distributions (Huang et al., 2023; Li et al., 2025a; Cai et al., 2026; He et al., 2025b). Related tasks estimate the proportion of watermarked text in a mixture or locate watermarked segments (Li et al., 2025b; Zhao et al., 2025; Li et al., 2024; Pan et al., 2025; Sander et al., 2026; Blanchet et al., 2026). Editing can weaken the evidence, motivating goodness-of-fit tests and analyses of robustness (Li et al., 2026; Krishna et al., 2023; Zhao et al., 2024; Golowich and Moitra, 2024).

We test for a watermark across the scored positions of a passage. The detector reconstructs Gumbel values, but their distribution under watermarked generation differs from that of Gumbel-max. Results about the power or optimality of a detector for Gumbel-max therefore do not transfer automatically. Section 3.2 gives our score and its null calibration under explicit independence assumptions.